

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
```

```
def predict(X, weights, lambd=1):
    net = np.dot(X, weights)
    return (1 / (1 + np.exp(-lambd * net)))

def calculate_dw(X, y, y_pred):
    return ((y_pred - y) * (y_pred) * (1 - y_pred) * X)

def calculate_error(yi, y_pred):
    return np.square(y_pred - yi)
```

```
def train_vanilla(X, y, weights, epochs=500, lr=0.001, epsilon=1e-8, beta=0.5):
    vel = 0
    for epoch in range(epochs):
        error = 0
        dw = 0
        for Xi, yi in zip(X, y):
            y_pred = predict(Xi, weights)
            dw += calculate_dw(Xi, yi, y_pred)
            error += calculate_error(yi, y_pred)

        vel = (beta * vel) + ((1 - beta) * (dw**2))
        weights -= ((lr * dw) / (np.sqrt(vel + epsilon)))
        error /= (2 * len(X))

        if (epoch+1) % 50 == 0:
            print(f'Weights after epoch {epoch+1} : ',weights)
            print(f'Error after epoch {epoch+1} : ',error)
```

```
def train_stochastic(X, y, weights, epochs=500, lr=0.001, epsilon=1e-8, beta=0.5):
    vel = 0
    for epoch in range(epochs):
        error = 0
        for Xi, yi in zip(X, y):
            y_pred = predict(Xi, weights)
            dw = calculate_dw(Xi, yi, y_pred)
            error += calculate_error(yi, y_pred)
            vel = (beta * vel) + ((1 - beta) * (dw**2))
            weights -= ((lr * dw) / (np.sqrt(vel + epsilon)))

        error /= (2 * len(X))

        if (epoch+1) % 50 == 0:
            print(f'Weights after epoch {epoch+1} : ',weights)
            print(f'Error after epoch {epoch+1} : ',error)
```

```
def train_mini_batch(X, y, weights, epochs=500, lr=0.001, epsilon=1e-8, beta=0.5, bs=32):
    vel = 0
    for epoch in range(epochs):
        error = 0
        dw = 0
        i = 0
        for Xi, yi in zip(X, y):
            y_pred = predict(Xi, weights)
            dw += calculate_dw(Xi, yi, y_pred)
            i += 1
            error += calculate_error(yi, y_pred)

        if i%bs == 0 or i == len(X):
            vel = (beta * vel) + ((1 - beta) * (dw**2))
            weights -= ((lr * dw) / (np.sqrt(vel + epsilon)))
            dw = 0

        error /= (2 * len(X))

        if (epoch+1) % 50 == 0:
            print(f'Weights after epoch {epoch+1} : ',weights)
            print(f'Error after epoch {epoch+1} : ',error)
```

```
df = pd.read_csv('bank_note.csv')
df.insert(4, 'x0', 1)
```

```
X = df[['variance','skewness','curtosis','entropy','x0']].values
y = df['class']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=13, test_size=0.2)
weights = input('Enter 4 weights and 1 bias : ').split()
weights = np.array([float(weight) for weight in weights], dtype='longdouble')
```

```
print('RMSPROP (Vanilla)')
train_vanilla(X_train,y_train,weights.copy())
```

```
➡ RMSPROP (Vanilla)
Weights after epoch 50 : [ 0.4493069 -0.25037206 0.74903664 -0.35157574 0.06198668]
Error after epoch 50 : 0.2163499406573366
Weights after epoch 100 : [ 0.39930191 -0.29993683 0.69883838 -0.40205345 0.11442909]
Error after epoch 100 : 0.20305518777499387
Weights after epoch 150 : [ 0.34930281 -0.34924393 0.64868769 -0.45234997 0.16504475]
Error after epoch 150 : 0.1900083577107779
Weights after epoch 200 : [ 0.29931734 -0.39733326 0.59856894 -0.50253783 0.21538433]
Error after epoch 200 : 0.1772488916715075
Weights after epoch 250 : [ 0.24933116 -0.40912908 0.54843979 -0.55255459 0.26544664]
Error after epoch 250 : 0.16454668072104164
Weights after epoch 300 : [ 0.19938231 -0.42414932 0.49835217 -0.60246006 0.31540519]
Error after epoch 300 : 0.1517611330275018
Weights after epoch 350 : [ 0.14946333 -0.4461929 0.44831095 -0.65229241 0.36532279]
Error after epoch 350 : 0.13943686248340303
Weights after epoch 400 : [ 0.09956499 -0.47276545 0.39830524 -0.7020503 0.41520511]
Error after epoch 400 : 0.12794796446157614
Weights after epoch 450 : [ 0.04968164 -0.50230417 0.34832318 -0.75171989 0.46505162]
Error after epoch 450 : 0.11750347212898721
Weights after epoch 500 : [-1.92201043e-04 -5.33548721e-01 2.98354188e-01 -8.01262652e-01
5.14854182e-01]
Error after epoch 500 : 0.10816936243524235
```

```
print('RMSProp (Stochastic)')
train_stochastic(X_train,y_train,weights.copy())
```

```
➡ RMSProp (Stochastic)
Weights after epoch 50 : [-2.86730008 -1.27913962 -1.75971618 0.38487542 4.40293968]
Error after epoch 50 : 0.005154824874154497
Weights after epoch 100 : [-3.76696051 -1.72619747 -2.38285566 0.49903036 5.67723716]
Error after epoch 100 : 0.005210018111249248
Weights after epoch 150 : [-4.5312659 -2.08855458 -2.85993662 0.54560109 6.48693452]
Error after epoch 150 : 0.005040569176895971
Weights after epoch 200 : [-5.22718951 -2.41013561 -3.28224936 0.5766594 7.12537978]
Error after epoch 200 : 0.004839186751381252
Weights after epoch 250 : [-5.93101704 -2.67360922 -3.66909961 0.62812689 7.66466291]
Error after epoch 250 : 0.004611839250624377
Weights after epoch 300 : [-6.64252837 -2.90169328 -4.0308283 0.71166195 8.15551326]
Error after epoch 300 : 0.004404283069751225
Weights after epoch 350 : [-7.34808213 -3.10411553 -4.37652138 0.80717939 8.61703642]
Error after epoch 350 : 0.0042898737027841525
Weights after epoch 400 : [-8.01688018 -3.29691396 -4.7078343 0.89389496 9.06299148]
Error after epoch 400 : 0.004289264429920012
Weights after epoch 450 : [-8.64235661 -3.4923312 -5.02952933 0.95850962 9.4899824 ]
Error after epoch 450 : 0.004333987126984063
Weights after epoch 500 : [-9.24463553 -3.70439112 -5.35502286 1.00025217 9.89914275]
Error after epoch 500 : 0.004364673110561608
```

```
print('RMSProp (Mini Batch)')
train_mini_batch(X_train,y_train,weights.copy())
```

```
➡ RMSProp (Mini Batch)
Weights after epoch 50 : [-0.90922785 -0.4857604 -0.5070907 -0.2269912 0.63426035]
Error after epoch 50 : 0.013924555020031978
Weights after epoch 100 : [-1.33849478 -0.70436909 -0.83546024 -0.0556999 1.48323007]
Error after epoch 100 : 0.006653368091379175
Weights after epoch 150 : [-1.61703277e+00 -8.45753275e-01 -1.02957262e+00 -1.39539789e-03
1.88733160e+00]
Error after epoch 150 : 0.005311757525720115
Weights after epoch 200 : [-1.83189912 -0.94747119 -1.1682011 0.01471029 2.14003183]
Error after epoch 200 : 0.004784596523900017
Weights after epoch 250 : [-2.01182614 -1.03352499 -1.28306475 0.01672463 2.3262747 ]
Error after epoch 250 : 0.00448527569604139
Weights after epoch 300 : [-2.17085422 -1.10825877 -1.38315843 0.01551631 2.47755462]
Error after epoch 300 : 0.004287034729684852
Weights after epoch 350 : [-2.31613325 -1.17447407 -1.47309208 0.01441242 2.60858229]
Error after epoch 350 : 0.004143551415122579
Weights after epoch 400 : [-2.45195354 -1.2344343 -1.55582696 0.01399621 2.7267326 ]
Error after epoch 400 : 0.0040334653462411425
Weights after epoch 450 : [-2.58093555 -1.28985074 -1.63333859 0.01409421 2.83596019]
Error after epoch 450 : 0.0039455612079365805
Weights after epoch 500 : [-2.70471446 -1.34195793 -1.70696586 0.01443457 2.93862241]
Error after epoch 500 : 0.003873355292103846
```